Gavin Tranquilino

Mechatronics Engineering Student

gavintranquilino.com
gtranqui@uwaterloo.ca
github.com/gavintranquilino
linkedin.com/in/gavintranquilino

Education

University of Waterloo **June 2028**
Candidate for BASc in Mechatronics Engineering *Waterloo, ON*

- **Coursework**: (DSA) Data Structures and Algorithms, Linear Algebra, OOP, FPGA, PLC, Materials, Statistics, Ordinary Differential Equations, RTOS STM32, Microprocessors, Actuators and Power Electronics, Design and Dynamics of Machines

Experience

Mechatronics Engineering Assistant January 2026 – April 2026
Acceleration Consortium *Toronto, ON*

- Led integration of a self-driving lab, automating a 2D gantry and 6DoF robot arm to run material discovery experiments.
- Deployed an **MQTT** control broker and bridged wired/wireless links for reliable realtime **IoT** messaging.
- Architected the networking stack to autonomously orchestrate 8 **IoT devices**, enabling system control over **ethernet switching**.
- Used **Python** to develop **inverse kinematics** for the robot arm to transport materials autonomously.
- CAD modeled parts in **Fusion360** for manufacturing and prototyping housings for electronics, and tooling fixtures.

Humanoid Robotics Engineering Co-op May 2025 – December 2025
WATonomous *Waterloo, ON*

- Building **robotic arms** (6DoF) with tendon driven anthropomorphic hands (20DoF each), aiming for VR teleoperation.
- Containerized ROS2 system in **Docker**, mounting CAN transceivers to enable communication between subsystems.
- Designed **URDF** models to define the transform **TF tree** for **RL simulation** and training in **NVIDIA Isaac Sim**.
- Built visualization infrastructure connecting **Gazebo** simulations to **Foxglove** for real-time debugging and data analysis.
- Assembled **PCBs** with 0.5mm pitch **SMD components**, soldered by hand, reducing assembly costs by 30%

Undergraduate Research Assistant September 2024 – December 2024
University of Waterloo Engineering IDEAs Clinic *Waterloo, ON*

- Designed a digital CAD twin of an existing knee crutch in **SolidWorks**.
- Instrumented a wearable knee crutch, allowing force readings for gait analysis and material selection via **FEA**.
- Developed a **data acquisition** system using **I2C** and C++, converting a bathroom scale for real-time load measurements.
- Built **Python** scripts for force distribution visualization in **Matplotlib**, with data logging for **gait analysis**.

Mechanical Engineering Associate January 2024 – April 2024
Sheartak Tools Ltd. *Waterloo, ON*

- Designed 15 third party woodworking machinery upgrades with **DFMA** in **SolidWorks** to meet OEM specifications.
- Applied **GD&T** principles to guarantee manufacturing accuracy for custom machine parts.
- Created 25 detailed installation manuals, including parts lists and assembly instructions, ensuring ease of use for customers.

Projects

Autonomous LiDAR Navigation for Mobile Robot

- Developed **C++ ROS2** nodes to convert **LiDAR** data into a **2D costmap** for obstacle detection and perception.
- Generated a **world model** from costmap and odometry data to represent the current environment.
- Implemented **A* algorithm** to compute obstacle-aware paths through the mapped environment.
- Applied **Pure Pursuit** to follow planned paths for smooth differential drive navigation.

Warehouse Autonomous Guided Vehicles (AGV)

- **Won TMMC Software Challenge** by developing autonomous warehouse robots using **TurtleBot 4** and **ROS2**.
- Generated a real-time costmap converting 2D **LiDAR** scans to **occupancy grids** with obstacle inflation for perception.
- Implemented **CV** stop sign detection using **YOLOv8** with bounding box distance estimation to stop at intersections.
- Designed **cascading PID controller** for wall-following and heading control with **IMU** feedback for warehouse traversal.
- Solved collision risks by implementing **LiDAR safety zones** with emergency stopping and backward movement protocols.

Technical Skills

Software/Languages: Python, C, C++, CMake, SSH, Bash, Gazebo, Foxglove, Linux, Ubuntu, JS, HTML, CSS, SQL, LaTeX
Libraries/Frameworks: ROS2, Docker, OpenCV, MQTT, Git, MediaPipe, Flask, NumPy, OpenGL
Mechanical: SolidWorks, Fusion360, AutoCAD, GD&T, CAD, FEA, DFMA, 3D Printing, Machine Tools, Onshape
Electrical: I2C, SPI, UART, CAN Bus, RS485, RS282, Arduino, Raspberry Pi, Soldering, Oscilloscope, LiDAR, PLC, HMI